Software Design Description 200

ESP32-DevKitV1 Main Controller – Overview

Overview

The ESP32-DevKitV1 is the central hub of the trail camera system. It receives raw image frames from the ESP32-CAM over SPI slave DMA, forwards them to the Tang Nano 9K FPGA over a second SPI master bus for WHT compression, manages peripheral sensors (temperature/humidity via I2C), drives a stepper motor for pan positioning (via FPGA I2C GPIO expander), and monitors three PIR motion detectors to trigger the capture pipeline. Future: receive compressed coefficients from the FPGA via SPI MISO and transmit them over LoRa UART to the S3 receiver.

> **Hardware:** ESP32-DevKitV1 | No PSRAM | Dual SPI buses (SPI2 slave, SPI3 master) | UART2 to LoRa | I2C0 to FPGA + AHT20

Detailed Documentation

SDD	Title	Covers
SDD201	Image Pipeline	cam_spi, image_processor, fpga_spi – full data path from camera to FPGA
SDD202	I2C Bus & FPGA GPIO	i2c_bus driver, bus recovery, PCA9534 expander, steppers, buttons
SDD203	Sensors, PIR & Motor	PIR detect-respond loop, AHT20 driver, stepper ramp profiles
SDD204	ISR, Utilities & Config	Semaphore dispatch, DMA alloc, wire protocol, build flags, init sequence
SDD205	Settings Display	Button-driven menu on TFT LCD via SPI pixel stream to FPGA BRAM – PLANNED
SDD010	LoRa Calculations	Link budget, spreading factor, bit rates
SDD011	LoRa Bench Test	Bench protocol, 5 bugs found, READY/START handshake

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Module Hierarchy

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main.c                          Entry point, init, task creation, heap monitor
|---- src/interprocessor/
|     |---- cam_spi.c           SPI slave -- receives frames from ESP32-CAM       [SDD201]
|     |---- fpga_spi.c          SPI master -- sends frames to FPGA                [SDD201]
|     |---- lora_uart.c         UART interface to RYLR998 LoRa module             [SDD010]
|     |---- lora_bench.c        LoRa bench test (TEST_MODE_LORA_BENCH)            [SDD011]
|     \---- i2c_bus.c           I2C master bus driver with recovery               [SDD202]
|---- src/peripherals/
|     |---- aht20.c             Temperature/humidity sensor (I2C 0x38)            [SDD203]
|     |---- fpga_gpio.c         FPGA I2C GPIO expander (I2C 0x27)                [SDD202]
|     |---- motor.c             Stepper motor: step table, ramp, swing            [SDD203]
|     |---- pir.c               PIR motion detector GPIO input                    [SDD203]
|     \---- sensors_temp_humidity.c Sensor task: PIR poll, AHT20, motor, enable   [SDD203]
\---- src/utilities/
      |---- image_processor.c   Camera-to-FPGA passthrough (future: ML, diff)     [SDD201]
      |---- isr.c               ISR semaphore init + handlers                     [SDD204]
      \---- utils.c             Checksum, DMA alloc, validation                   [SDD204]
```

Operating Modes

Production Mode (default)

Five FreeRTOS tasks run concurrently:

Task	Function	Priority	Stack	Purpose
cam_rx	`cam_spi_receive_task`	High (5)	4 KB	Receive frames from ESP32-CAM over SPI
img_proc	`image_processor_task`	Medium (4)	8 KB	Dequeue frames, forward to FPGA
lora_rx	`lora_receive_task`	Medium (4)	4 KB	Listen for LoRa commands from S3
sensors	`sensors_task`	Low (3)	4 KB	PIR polling, AHT20, motor control
(main)	`monitor_heap`	–	–	Heap usage logging every 10s

LoRa Bench Mode (`TEST_MODE_LORA_BENCH`)

Only the `lora_bench_task` runs. All camera, FPGA, and sensor initialization is skipped. See SDD011.

Other Test Modes

- `TEST_MODE_FPGA_PATTERNS` – cycle SPI test patterns to FPGA on button press
- `TEST_MODE_FPGA_GPIO` – exercise I2C GPIO expander (buttons + steppers)
- `TEST_MODE_LORA_LOOPBACK` – local LoRa echo
- `TEST_MODE_CAMERA_INJECT` – inject synthetic frames at interval
- Mutually exclusive (enforced by `#error` in `config.h`)

Bus Summary

Bus	Host	Role	Peripheral	Clock	Detail
SPI2	CAM_SPI_HOST	Slave	ESP32-CAM	10 MHz (master sets)	DMA, 4 KB chunks
SPI3	FPGA_SPI_HOST	Master	Tang Nano 9K	9 MHz	DMA, 32 KB max, async queue/get
UART2	LORA_UART_NUM	–	RYLR998 LoRa	115200 baud	AT command protocol, 8N1
I2C0	FPGA_I2C_PORT	Master	AHT20 + FPGA GPIO	100 kHz	Mutex-protected, auto-recovery

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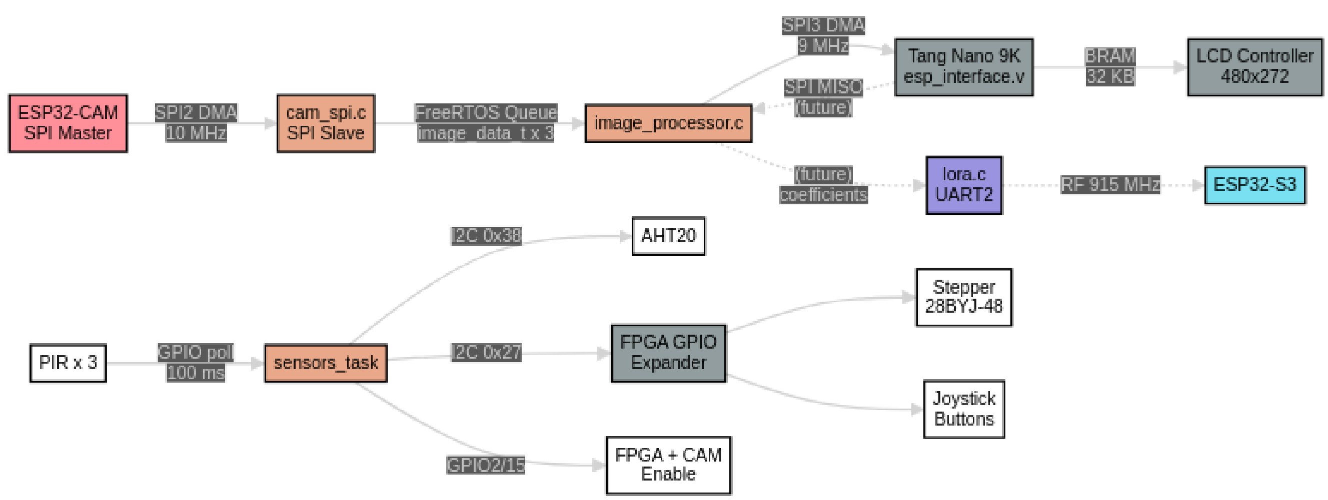
Pin Map

GPIO	Function	Direction	Bus/Module
23	SPI MOSI (cam)	In	SPI2 slave
19	SPI MISO (cam)	Out	SPI2 slave
18	SPI SCLK (cam)	In	SPI2 slave
5	SPI CS (cam)	In	SPI2 slave
32	SPI MOSI (fpga)	Out	SPI3 master
33	SPI MISO (fpga)	In	SPI3 master

GPIO	Function	Direction	Bus/Module
25	SPI SCLK (fpga)	Out	SPI3 master
26	SPI CS (fpga)	Out	SPI3 master
17	UART TX (lora)	Out	UART2
16	UART RX (lora)	In	UART2
21	I2C SDA	Bidir	I2C0
22	I2C SCL	Out	I2C0
35	PIR1	In	sensors
34	PIR2	In	sensors
4	PIR3	In	sensors
13	Stepper IN1	Out	motor
12	Stepper IN2	Out	motor
14	Stepper IN3	Out	motor
27	Stepper IN4	Out	motor
2	FPGA enable	Out	sensors
15	Camera enable	Out	sensors
0	Test button	In	ISR (GPIO interrupt)

Data Flow Summary



References

- SDD000 – System Overview
- SDD100 – ESP32-CAM
- SDD300 – ESP32-S3 Receiver
- SDD400 – FPGA Subsystem